Homework 6

Problem 1. Prove the constant rank theorem.

Let $f : U \to \mathbb{R}^n$ be C^1, where $U \subseteq \mathbb{R}^d$ is open and f' has constant rank r near point $p \in U$. Show that we can find local C^1 diffeomorphisms ϕ and ψ such that

$$\psi \circ f \circ \phi \left(\widetilde{x_1}, \ldots, \widetilde{x_d}\right) = \left(\widetilde{x_{d-r+1}}, ..., \widetilde{x_d}, \underbrace{0, \ldots, 0}_{n-r \text{ zeros}}\right)$$

for all $\left(\widetilde{x_1}, \ldots, \widetilde{x_d}\right)$ near $\phi^{-1}(p)$.

Hint: WLOG try to split $f = (f_1, f_0)$ where $f_1 : U \to \mathbb{R}^r$ and $f_0 : U \to \mathbb{R}^{n-r}$ such that f_1' is surjective on U.

Problem 2. Let $E \subseteq \mathbb{R}^d$ be a closed set and $f : E \to E$ continuous such that

$$|f(x) - f(y)| \leq \alpha |x - y|$$

for some $\alpha \in (0, 1)$. Show that there is a <u>unique</u> fixed point $p \in E$ such that $f(p) = p$.
Hint: pick any starting point p and look at $(f^n(p))_{n \in \mathbb{N}}$.

Problem 3. Show the 2nd derivative test.

For a symmetric matrix A, we write $A > 0$ when $\langle Ax, x\rangle > 0 \; \forall x \neq 0$ and $A \geq 0$ when $\langle Ax, x\rangle \geq 0 \; \forall x \neq 0$ (by the spectral theorem, these are equivalent to all eigenvalues being positive and non-negative respectively).

Let $f : U \to \mathbb{R}$ be C^2 where $U \subseteq \mathbb{R}^d$ is open. Then $f''(p)$ is the (symmetric) Hessian matrix at p.

1. If $p \in U$ is a local minimum point then $f'(p) = 0$ and $f''(p) \geq 0$.

2. If at $p \in U$ we have $f'(p) = 0$ and $f''(p) > 0$ then p is a local minimum point of f.